

ME 419 Midterm Report: Simulations in Moldex 3D

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Introduction

Injection molding is the second-highest polymer manufacturing process after extrusion molding, due to its high manufacturability, mass production capabilities, and general ease of use. However, the difficulty lies in creating the proper mold and in-mold conditions that allow for good part quality, while not sacrificing production time and profit. The purpose of this project is to gain a better understanding of injection molding principles, as well as design a runner and gate system with a cooling channel to effectively mold a part. Using simulation tools such as Moldex 3D, models of temperature, warpage, and packing can be made and optimized to find the best solution.

This report will model a Pi-shaped cookie cutter found on Thingiverse [1]. First, the model mesh is edited slightly, and a material is chosen based on its use requirements. Initial testing process conditions were found through research and calculations, and were set to model four different filling methods with runner and gate systems. The best overall design was selected, and the process was repeated for three different-shaped cooling channels. Lastly, the input data was adjusted slightly to create the most optimal process. The report will discuss in further detail the manufacturability of the design, along with any environmental impacts production could incur.

Input Data

Part Design

The part chosen to be modeled is a cookie-cutter in the shape of the mathematical symbol pi, shown by Figure 1.

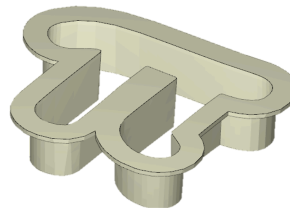


Figure 1. Pi cookie cutter

This piece can be difficult to injection mold due to its non-uniformity and multi-wall thicknesses, but as the project's goal is to gain a better understanding of injection molding, a challenging part will allow for more thorough analysis. Since this piece is incredibly thin, warping is also expected. A lateral warpage that maintains a recognizable pi shape will be okay for use, as long as there is minimal vertical warpage that could affect its cookie-cutting abilities. For use, the part must be food-safe and reusable, so it may be used for many batches of cookies.

The final mold size is 150mm x 150mm x 150mm. This was chosen as it is the smallest mold that can fully fit the part, the runners and gates, and the cooling system. Based on the simulations, it will be determined whether a two or three-part mold is necessary.

Materials

Material selection was mainly influenced by its use. The plastic must be completely solid when finished, and most importantly, must be food-safe. It would also be beneficial to have the part be reusable, with good chemical resistance, and recyclable when no longer needed. Considered were Polypropylene and High-Density Polyethylene (HDPE), both common materials when used for food-related items. Their attributes are listed in Table 1 and were found through material research [2],[3],[4],[5].

Table 1. Material Comparison between Polypropylene and High Density Polyethylene

	Polypropylene (PP)	High Density Polyethylene (HDPE)
Melting Temperature [°C]	210-290	180-280
Chemical Resistance	Good to acids and alcohols, poor to hydrocarbons	Good to acids and alcohols, poor to hydrocarbons
Food Safety	Food Safe	Food Safe
Estimated Cost [\$ /kgram]	\$1.11	\$0.95
Recyclability	Recyclable, but difficult	100% recyclable
Typical Shrinkage	1.5%	1.5%
Other factors	Low melt viscosity	High injection speed
Packing pressure	80 MPa recommended	70-105 MPa recommended

After comparison, HDPE was chosen for its recognizable lower cost and higher recyclability. It also offers slightly lower melt temperatures and packing pressure, decreasing the overall cost of running an injection molding machine. On Moldex3D, the specific material chosen was HDPE Borealis PE MG75475S Borealis, and its properties are shown in the graphs below. These graphs show how pressure affects specific volume, which was taken into account when creating the initial process conditions.

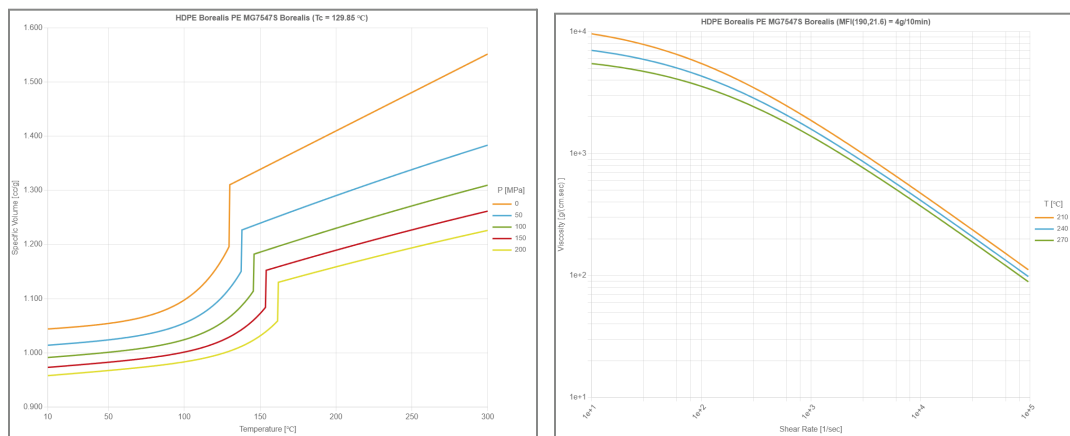


Figure 2. PvT and viscosity graphs of HDPE Borealis PE MG75475S Borealis

Initial Process Conditions

From this material, along with other research, initial process conditions were selected, which will be used for all simulation trials. Having the same process conditions for each trial will allow for a better comparison between designs, and once the best design is chosen, the process conditions can be optimized. These process conditions are listed in Table 2 and were used to help determine the runner and gate system and the cooling channels.

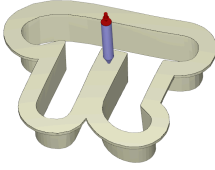
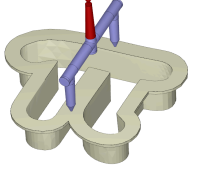
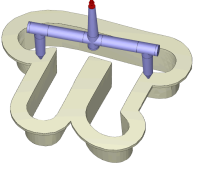
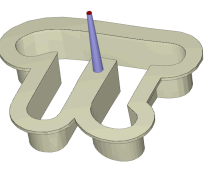
Table 2. Initial Process Parameters

Max injection pressure [MPa]	100
Max packing pressure [MPa]	92
Filling time [s]	.34
Filling percentage [%]	98
Packing time [s]	3
Melt temperature [°C]	215
Mold temperature [°C]	20
Air temperature [°C]	25
Cooling time [s]	10.6
Mold open time [s]	5
Final Cycle time [s]	18.94

Runner and Gate Systems

Four different simulations were run to determine the best gate and runner design. They were run without cooling channels to act as a ‘base’ condition, and all used the same initial process parameters shown in Table 2. The size of runners and gates was all found to be optimal according to the C-Mold Design Guide [6]. The four simulations are shown in Table 3.

Table 3. Gate Design Simulations

	Simulation 1	Simulation 2	Simulation 3	Simulation 4
Gate/ Sprue Design				
Pros	Easy design, central mold point	Smaller pin gates, little surface defect	Smaller pin gate, little surface defect	Easy design, central mold point
Cons	Could cause air traps or short shot	3 mold plates, weld lines likely where melt meets	3 mold plates, weld lines likely where melt meets	Could cause air traps
Size of runners	Diameter = 4 mm Height = 20 mm	Diameter = 4 mm Height = 20 mm	Diameter = 4mm Height = 20 mm	N/A
Size of gates	D_min = 1 mm D_max = 4 mm Length = 4 mm	D_min = 1 mm D_max = 4 mm Length = 4 mm	D_min = 1 mm D_max = 4 mm Length = 4 mm	D_min = 1 mm D_max = 4 mm Length = 30 mm

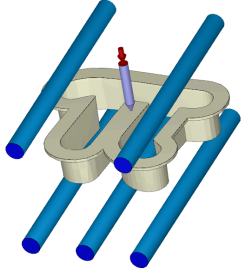
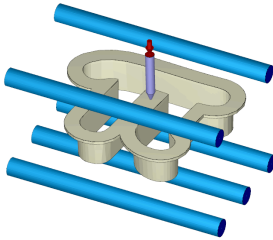
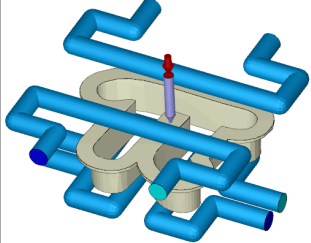
The size of the gates is extremely important in injection molding. The pin gates allow easy disconnection from the part with few surface defects. Gate location will help ensure even filling and packing, but having multiple gates could cause weld lines. A larger gate will cause a lower flow rate, and possibly cause the melt front to cool before it fully fills the cavity. Gate locations are also likely areas for sink marks, as the larger

volume of material cools in the gate and pulls up material from the part. In this simulation, sink marks will not affect the final part quality very much, and the pin gates are small enough to neglect most sink marks.

Cooling Channels

After a gate system was chosen, three different cooling channel configurations were tested. The goal of this was to provide even cooling while decreasing warpage. Uniform cooling is incredibly important in injection molding. Having non-uniform cooling will cause the melt in some areas of the mold to solidify, while the rest is still liquid. This imbalance of stresses will appear as sink marks and weldlines on the part surface or can cause warpage. Table 4 shows the design and reasoning behind the simulations. All channels have a diameter of 8 mm.

Table 4. Cooling Channel Simulations

	Simulation A	Simulation B	Simulation C
Cooling Channel Design			
Pros	Easy to machine	Easy to machine	Covers more of part
Cons	Misses center of part	Misses center of part	Difficult to machine

When designing the cooling channels, it was vital that they evenly covered as much of the part surface as possible, especially the center. Since the center of the part is where the gate is, it will be the hottest region, and take the longest to cool. Simulations A and B will help show if there is a difference in the direction of the cooling channels, and Simulation C shows channels that more reflect the part geometry.

Final process conditions

The time for each process depends mostly on the cooling and filling times. Filling time is optimized right when the gate freeze-off occurs, ensuring short shot does not occur, but the part is not overpacked. The optimized fill time can be solved through experimentation, finding the point where mass increase caps off. Cooling time is also an important factor, taking up over 70% of the cycle time in most molding cycles. Having too much cooling will create a less-than-optimal production rate, while having too little will cause the part to warp. A good volumetric flow rate is also important for filling the cavity, working mainly with hold pressure and melt temperature, allowing the viscous melt to fill the mold. Having too high a pressure will cause flash issues on the final part, and too low a pressure will lead to short shot.

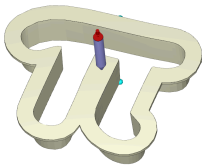
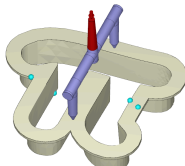
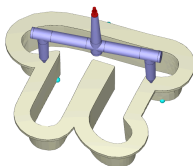
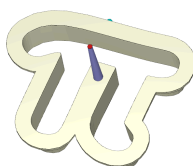
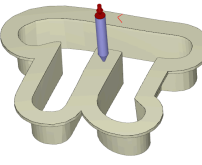
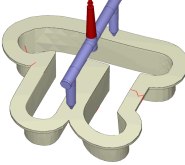
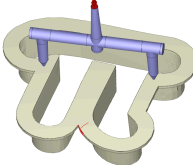
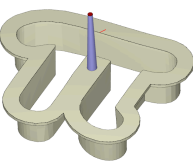
These were optimized by running many trials with slightly adjusted values, and looking at the final part temperature, density, air traps, weld lines, and warpage. If there were issues with any of these, process conditions were changed accordingly. For example, to fix weldlines, the melt temperature was increased slightly, and for warpage, the cooling time was increased. A table of some of these trials is in the Appendix.

Results and Discussion

Optimal Simulation Results

Starting with the gate design, the two main factors looked at were air traps and weld lines. Table 5 shows this comparison for each simulation. Other factors are shown in the Appendix, but were not heavily considered due to their similarity. Note in Simulation 4, there are also a few weld lines across the bottom of the part.

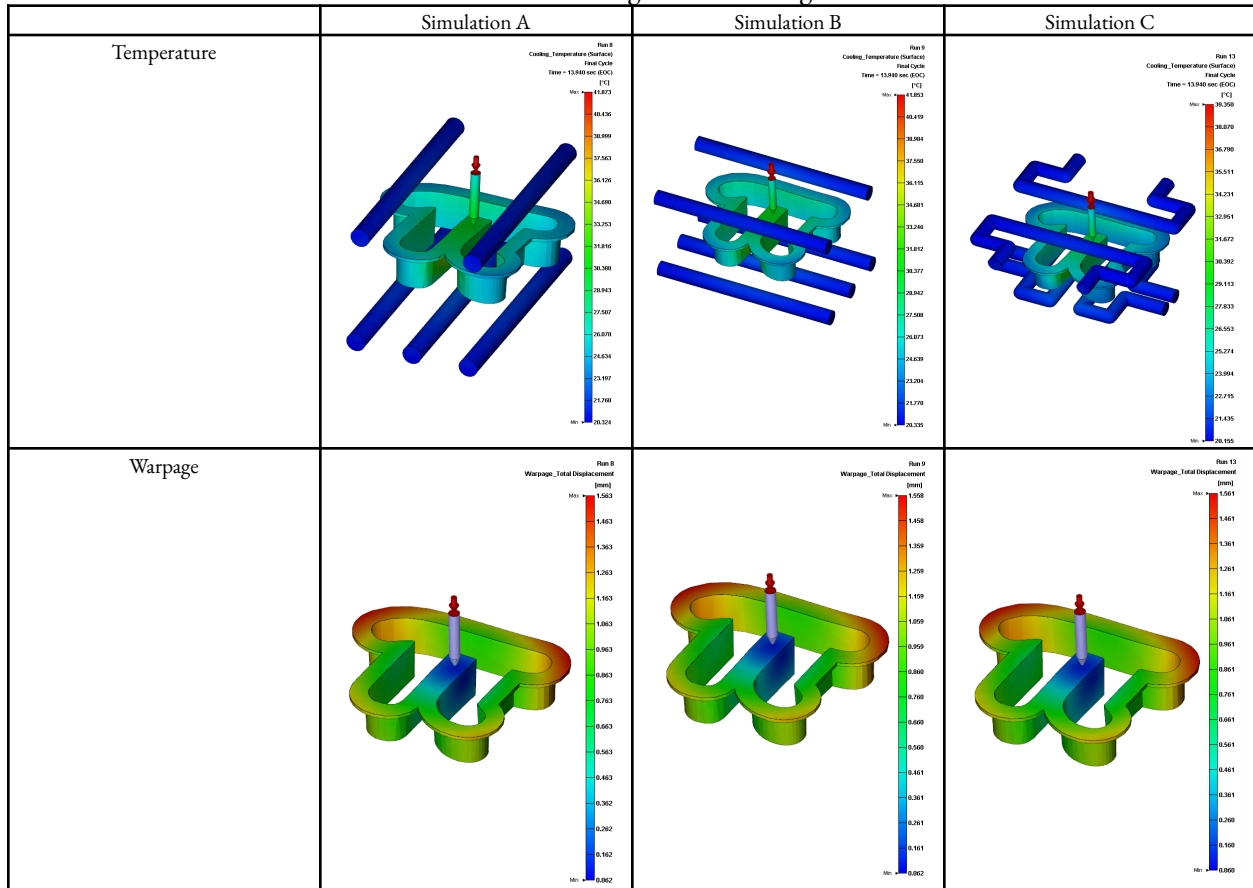
Table 5. Results from Gate Design Simulations

	Simulation 1	Simulation 2	Simulation 3	Simulation 4
Air Traps				
Weld Lines				

This analysis showed that Simulation 1 was the best. This simulation had the fewest air traps and weld lines, and a comparable density and heat distribution to the other tests. The weld lines that were there still maintained 100% strength as the rest of the part, so they could practically be ignored. Due to the ease of geometry, the mold only requires two plates, saving money during mold manufacturing.

Next, moving on to the cooling channels, the temperature distributions and warpage were the main results looked at, as seen in Table 6.

Table 6. Results from Cooling Channel Design Simulation



From the data, Simulation C had the lowest temperature values by about 2 degrees, but Simulation B ended up having the smallest warpage by about .02 mm. Although the end gave very similar results and either simulation would be a feasible choice, Simulation B was selected for its ease of machining, saving thousands of dollars when creating the mold, and giving desirable results.

Lastly, the process parameters were changed to create optimal results. The final parameters are in Table 7 below.

Table 7. Final Process Parameters

Max injection Pressure [MPa]	180
Max Packing Pressure [MPa]	100
Filling time [s]	.25
Packing time [s]	3
Melt temperature [°C]	210
Mold temperature [°C]	15
Air temp [°C]	25
Cooling time [s]	14
Mold open time [s]	1
Final Cycle time [s]	18.25

The final part should be free of sink marks, warpage, weldlines, and air traps. Although it is impossible to completely evade all of these, it is possible to change the process conditions to limit them. By changing the process parameters, all weld lines disappeared, along with one of the air traps. Since the other air trap is at the

edge of the part, it can be easily vented, possibly with an ejector pin nearby, and will not impact the final product. Lastly, sink marks and warpage can be controlled through holding and packing time, and an optimal cooling system. As discussed before, lateral warpage does not matter too much for the part, as long as the general shape is kept and recognizable. Since the starting part was extremely thin, a high warpage is expected. In the end, the maximum warpage was 1.452 mm, and the part still maintained its shape and usability, so the simulations proved successful.

It is also worth mentioning the manufacturability of the mold itself. The simple runner design allows for only two plates in the mold, creating an easy path for the melt to travel. The chosen cooling channels are also easy to build, as they are simple, straight channels running all the way through the mold. Even if the design in simulation C was chosen, new manufacturing techniques like 3D printing would still allow for the mold to be made; it would just be considerably more expensive.

For production, a manufacturer would likely create a mold with more than one cookie-cutter cavity, possibly 4 or 6. As this part likely has low demand, it would make sense to have this mold have fewer cycles with more cavities, resulting in shorter processing and production time.

To determine the feasibility of production, cost calculations were carried out. As seen in the Appendix, if production lasts 100 hours/week, the material needed will cost \$473.42. Selling each part for \$4.99 and estimating around \$3,500/week for energy and production costs, the net profit each week would be \$92,964.21. Start-up costs for injection molding machines are incredibly high, both for the machine itself and the creation of the mold. However, with the high profit each week, these start-up costs should not be an issue, and it would make sense to manufacture the cookie-cutter.

The environmental friendliness of injection molding is based mainly on two factors: energy and materials. Injection molding machines use a lot of energy, with older hydraulic oil machines making up 20% of the world's carbon emissions [7]. More recently, electric machines have been implemented, replacing the use of fossil fuels with cleaner energy, and allowing for easier changes to process parameters.

Many materials used in injection molding are not biodegradable and can live in landfills for hundreds of years. The best way to combat this is by using fully recyclable material, such as the HDPE in this design. Recyclable material will greatly decrease pollution, as well as conserve resources. So, for this method, the recyclability of the material helps diminish the carbon footprint and makes the process more eco-friendly. Moreover, with a recycling code #2, HDPE is one of the easiest materials to recycle, with almost 30% of all HDPE bottles recycled [8]. On the other hand, HDPE is made from fossil fuels and is not biodegradable, thereby harming the environment. To make this part eco-friendly, there is a dependence on the population using the cookie-cutter, and the hope that they will recycle it to be melted down for future use.

Summary

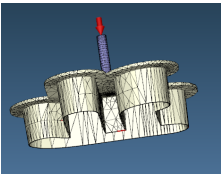
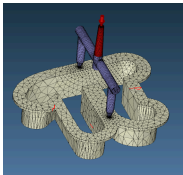
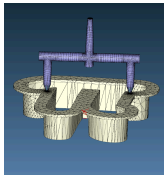
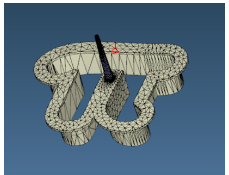
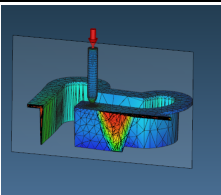
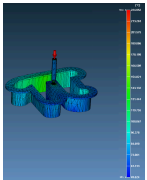
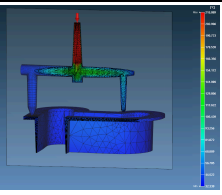
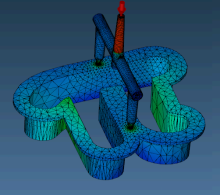
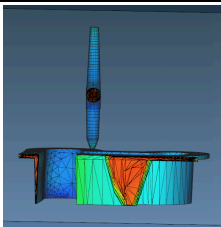
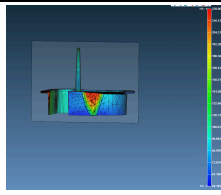
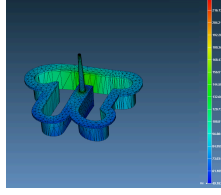
Based on the analyzed model of a pi shaped cookie-cutter, an optimal gate design of a single pin gate in the center of the mold was chosen due to its ease of manufacturing, and limited air traps and weld lines. Different cooling channels were designed and analyzed for part temperature and warpage. Straight cooling channels shown in Simulation B yielded similar results to the curved ones in Simulation C, but Simulation B was chosen for less warpage and easier mold manufacturing. HDPE proved to be a cheap, food-safe material to use but its high shrinkage, paired with the thin walls on the part, did cause noticeable warpage, even after process conditions were optimized. In the end, the part and mold is feasible to manufacture in a real-world setting, and could make a high profit, all while helping the environment with the highly recyclable material.

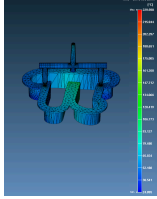
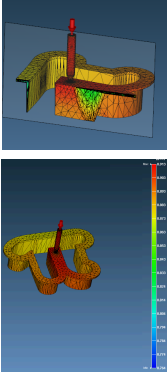
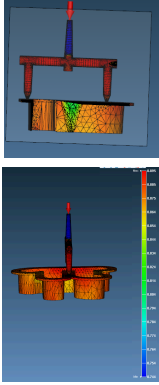
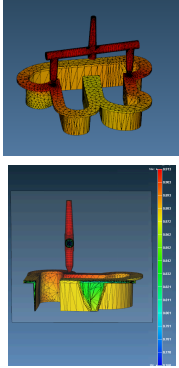
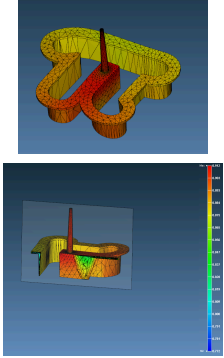
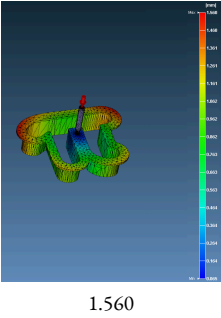
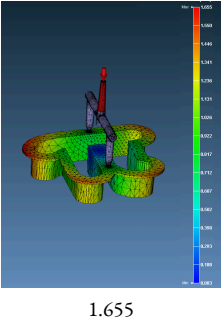
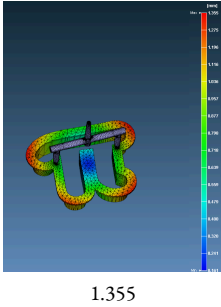
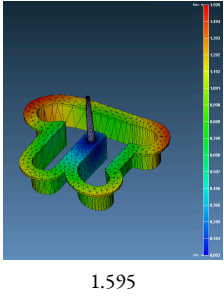
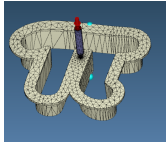
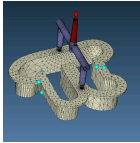
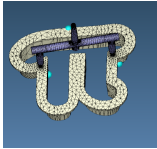
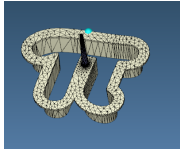
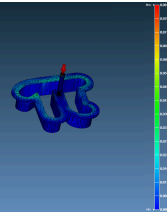
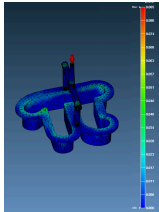
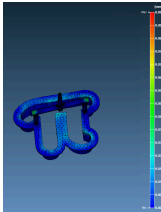
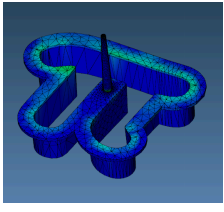
References

- [1] Thingiverse.com. "Pi Cookie Cutter - and Cookie Cutter Script." Thingiverse, 2025, www.thingiverse.com/thing:28882. Accessed 5 Nov. 2025.
- [2] Hindle, Colin. "Polypropylene (PP)." BPF, British Plastics Federation, 2019, www.bpf.co.uk/plastipedia/polymers/pp.aspx.
- [3] British Plastics Federation. "Polyethylene (High Density) HDPE." Bpf.co.uk, 2013, www.bpf.co.uk/plastipedia/polymers/hdpe.aspx.
- [4] "HDPE Injection Molding Processing Condition." Wwww.viewmold.com, www.viewmold.com/sources/Plastic_resin_injection_molding_processing_condition/HDPE_injection_molding_processing_condition.html.
- [5] Mike. "HDPE Price Index." Businessanalytiq, 3 Sept. 2020, businessanalytiq.com/procurementanalytics/index/hdpe-price-index/.
- [6] AC Technology. (1996). C-mold Design Guide: A Resource for Plastics Engineers. AC Technology.
- [7] ESSENTRA COMPONENTS. "Is Injection Moulding Environmentally Friendly?" Wwww.essentracomponents.com, 17 Mar. 2023, www.essentracomponents.com/en-gb/news/manufacturing/injection-moulding/is-injection-moulding-environmentally-friendly.
- [8] Plastics For Change. "Which Plastic Can Be Recycled?" Plastics for Change, 20 May 2021, www.plasticsforchange.org/blog/which-plastic-can-be-recycled.
- [9] "Pricing Your Work: The Machine Rate Enigma." Plasticstoday.com, 2023, www.plasticstoday.com/resin-pricing/pricing-your-work-the-machine-rate-enigma.

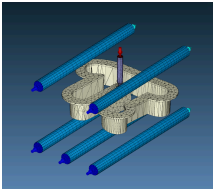
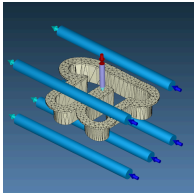
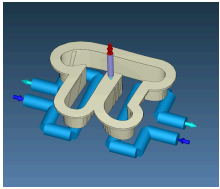
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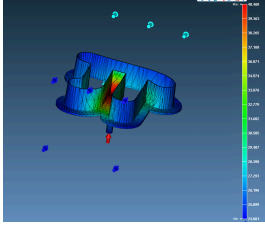
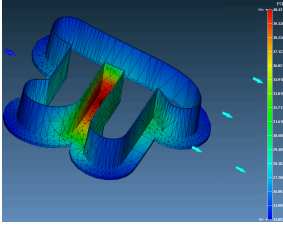
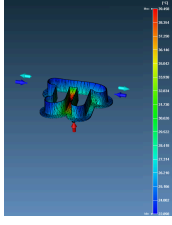
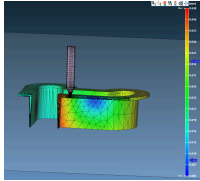
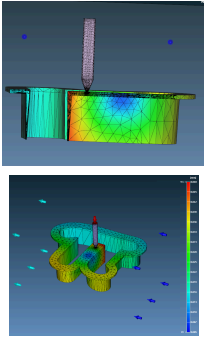
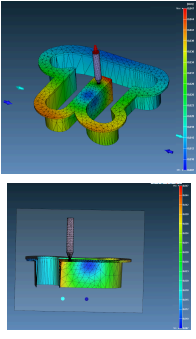
Gate/Runner Design

	Simulation 1	Simulation 2	Simulation 3	Simulation 4
Filling time	.351	.353	.348	.350
Weld lines				
				weld lines across bottom
Temperature distribution	 	 		 

				
Packing density distribution				
Packing time	3.364	3.378	3.343	3.395
Max warpage	 1.560	 1.655	 1.355	 1.595
Air Traps				
Sink Marks				

Cooling Channel Design

	Simulation 1	Simulation 2	Simulation 3
Channel Design			
Max Temperature [C]	103.666	105.486	105.688

Uniformity	 40.46	 40.431	 39.45
Warpage			

Tested Process Parameters

	Max injection Pressure [MPa]	Max Packing Pressure [MPa]	Filling time [s]	Packing time [s]	Melt temperature [C]	Mold Temperature [C]	Air Temperature [C]	Cooling time [s]	Mold Open time [s]	Final Cycle time [s]
Run 1	100	92	.34	3	215	20	25	10.6	5	18.94
Run 2	150	105	.5	3	250	25	25	10.6	3	17.1
Run 3	150	105	.3	5	210	20	20	12	2	19.3
Run 4	100	100	.25	3	240	15	20	13	1	17.25
Run 5	110	92	.25	3	210	15	25	14	1.5	18.75
Run 6	180	100	.25	3	210	15	25	14	1	18.25

Cost of production

Time = 100 hours/week → 360000 s /week

\$0.95/kg material

Part + sprue & gate 25.47 g = \$0.024

Assume 2% scrap

Production time 18.25 s

(time per week)/(production time)*.024 → cost of production per week = \$473.42

Sell for \$4.99 - comparable market prices

(time per week)/(production time)*.98*4.99 → profit per week = \$96,464.21

Cost of operation for 1 week = \$3,500

Net gain from 1 week = \$92,964.21